Integration By Parts

$\boxed{\text{Recall:}}$ For $y = u(x) \cdot v(x) \Longrightarrow y' = u'v + uv'$

$\boxed{\text{Theorem}}$ $\displaystyle\int u'(x)\,v(x) + u(x)\,v'(x)\,dx = u(x)\,v(x) + C$

or $\displaystyle\int u'v + uv'\,dx = uv + C$

$\boxed{\text{Theorem}}$ Formula for Integration by Parts.

For u, v differentiable,

$$\int u\,dv = uv - \int v\,du$$

$\boxed{\text{Note}}$ Saw: $u = x^2 \Longrightarrow du = 2x\,dx$

$u = \quad \Longleftarrow \quad du = 3x^2\,dx$

$v = \quad \Longleftarrow \quad dv = \cos(x)\,dx$

$\boxed{\text{Strategy}}$ $\displaystyle\int x e^{x^2} dx, \int x^2 e^x dx, \int x e^x dx, \int e^{x^2} dx$

$\boxed{\text{Formulae}}$ (1) $\int u\,dv = uv - \int v\,du$

(2) $\int_a^b u\,dv = uv\Big|_{x=a}^{x=b} - \int_a^b v\,du$

$\boxed{\text{Example 1}}$ Integrate by parts

(1) $\int x e^x\,dx$

(2) $\int x^2 \ln(x)\,dx$

$\boxed{\text{Example 2}}$ Integrate by parts

(1) $\int x^2 e^x \, dx$

(2) $\int e^x \sin(x) \, dx$

3.1

Evaluate the integral below.

$$\int \sin(\ln(5x))\,dx$$

Be sure to place the argument of any trigonometric or logarithmic functions in parentheses in your answer.

Trigonometric Integrals

$\boxed{\text{Theorem}}$ $1 - \cot(x) = \csc^2(x)$

$\sin^2(x) + \cos^2(x) = 1$, $\tan^2(x) + 1 = \sec^2(x)$

$\boxed{\text{Example 1}}$ Solve.

(a) $\int \cos^5(x)\,dx$

(b) $\int \sin^8(x)\cos^3(x)\,dx$

(c) $\int \sin^3(x)\cos^4(x)\,dx$

$\boxed{\text{Theorem}}$ Double-Angle Formulae

(a) $\sin^2(x) = \frac{1}{2}(1-\cos(2x))$ (b) $\cos^2(x) = \frac{1}{2}(1+\cos(2x))$

$\boxed{\text{Example 2}}$ Solve.

(a) $\int \sin^2(x)\,dx$

(b) $\int \cos^4(x)\,dx$

$\boxed{\text{Note:}}$ $\cosh^2(x) - \sinh^2(x) = 1$